

Anshuman Biswas

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EDUCATION

Indian Institute of Information Technology, Kottayam

Bachelor of Technology in Computer Science and Engineering (CGPA: 9.08 / 10.0)

Kottayam, Kerala, India

Aug. 2024 – May 2028

Delhi Public School, Sushant Lok

Class XII (CBSE): 94.2% | Class X (CBSE): 97.2%

Gurgaon, Haryana, India

Apr. 2021 – May 2024

TECHNICAL SKILLS

Languages: C++, Python, TypeScript, JavaScript, SQL (PostgreSQL, MySQL), C, HTML/CSS

AI & Agent Architecture: LangGraph, LangChain, DeepAgents, Multi-Agent Systems, LLM Evaluation, RAG

Distributed Systems & Storage: LSM Trees, Write-Ahead Logs (WAL), Bloom Filters, Apache Kafka, Debezium, Redis, Elasticsearch

Backend & Cloud: FastAPI, Node.js, Express, REST APIs, WebSockets, SSE, Supabase, Cloudflare (R2/D1), GCP, Render

Developer Tools & Collaboration: Docker, Kubernetes, Linux, Git, GitHub Actions, CMake, Google Benchmark, ngrok, Jira, Slack

Spoken Languages: English (Fluent), Hindi (Fluent), Bengali (Native), French (Elementary)

EXPERIENCE

AI Engineering Intern

Oct. 2025 – Jul. 2026

theMonks.tech

Los Angeles, CA (Remote)

- Architected specialized multi-agent workflows with persistent memory, domain-specific execution skills, and user clarification loops to autonomously parse, transform, and plot complex datasets.
- Refactored sequential agent pipelines into dependency-aware DAGs, parallelizing independent tasks to achieve a **2× speedup** and cut overall response times from **30s to sub-15s**.
- Optimized container microservices to slash memory consumption by **80% (from ~250MB to ~50MB)**; benchmarked server deployments via **ngrok** tunnels, reducing cross-server latency by **85% (from 1.4s to 0.2s)**.
- Designed end-to-end system and application logging architectures; monitored live deployments and real-time telemetry across **GCP** and **Render** to track resource utilization, token budgets, and container health.

Marketing & Legal Intern

May 2026 – Present

ELSTRAT

Remote

- Conducted comprehensive market research and competitive analysis for early-stage technology startups across the Indian startup ecosystem.
- Assisted founding teams with legal documentation, regulatory compliance, and due diligence frameworks.

PROJECTS

TransactFlow | C++17, PostgreSQL, Apache Kafka, Debezium, Redis, Elasticsearch, Docker

Sept. 2025

- Engineered an enterprise Change Data Capture (CDC) streaming pipeline ingesting **1,000+ req/s** into PostgreSQL, decoding WAL with Debezium and replicating to dual sinks with **zero data loss** over 216k+ soak events.
- Developed high-throughput C++ consumer workers featuring manual Kafka offset commits, exponential backoff retries, and Dead-Letter Queue (DLQ) isolation for poisonous payloads.
- Guaranteed sub-millisecond cache consistency by implementing atomic Lua-scripted Log Sequence Number (LSN) monotonicity guards in Redis.

FlashDB | C++20, CMake, Google Benchmark, Python

Aug. 2025

- Designed and implemented an embedded Log-Structured Merge (LSM) tree key-value storage engine in C++20 with decoupled Write-Ahead Log (WAL), MemTable, SSTables, and Size-Tiered Compaction.
- Integrated probabilistic **Bloom Filters** to eliminate redundant disk probes, achieving an **~85× read speedup** for non-existent key lookups under Google Benchmark stress testing.

DeepTrade | Python, FastAPI, LangGraph, DeepAgents, Upstox API, Supabase, Telegram API

July 2025

- Architected a self-hosted agentic trading assistant coordinating specialized research and broker-execution subagents for real-time market queries and live/paper order execution on Upstox via Telegram with TOTP two-factor security.
- Investigated and upstream-reported two critical bugs in the official **Upstox Python SDK** (#155, #156) regarding singleton configuration state leakage and urllib3 ≥ 2.0.0 deprecations.

CERTIFICATIONS

Anthropic: Claude with the Anthropic API — Hands-on LLM integration, context-aware architectures, and agentic workflows (Cert: s2xodzvren54)

LangChain Academy: Project: Deep Agents — Building autonomous, long-running AI agents on LangGraph and LangChain (Cert ID: wtfjoejcfo)

Oracle University: Oracle Cloud Infrastructure (OCI) 2025 Certified AI Foundations Associate (Cert: 3231424800CI25AICFA)

ROLES OF RESPONSIBILITY

BetaLabs (Official Technology Club of IIIT Kottayam)

Kottayam, Kerala, India

Lead – App Development Club | Sublead – AI & Data Science Club

Aug. 2025 – Present

- **Lead, App Dev Club (Aug. 2026 – Present):** Core organizer and contributor for annual Techfest "Aaroh" with 300+ participants; managed hackathons, technical workshops, and hands-on coding sessions primarily for freshmen and sophomore students.
- **Sublead, AI & DS Club (Aug. 2025 – Aug. 2026):** Conducted interactive hands-on sessions on foundational AI/ML concepts and volunteered in university-wide hackathons to mentor student development teams.